Seasonal Source Water Temperature as a Predictive Indicator for Legionella Outbreak Risk in Ontario Healthcare Facilities: A Correlation Analysis Using PWQMN Baseline Data and Provincial Surveillance Records 2019–2025

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Related Work

This study extends the Presignal Subtraction Framework first applied in: Carter, J. (2026). Chronic Source Water Quality Exceedances in the Grand River Watershed: Implications for AI Data Center Cooling Infrastructure in Ontario, Canada. Zenodo. DOI: 10.5281/zenodo.19653415

Data Availability

Ontario PWQMN data: data.ontario.ca and DataStream (DOI: 10.25976/tnw0-3x43). PHO legionellosis surveillance reports: publichealthontario.ca. Presignal statistical baseline available from the corresponding author upon request. AquaSignal dashboard: www.ufosworldwide.com/aquasignal/

Conflict of Interest

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Abstract

Legionellosis — the spectrum of illness caused by Legionella bacterial infection — follows a pronounced and consistent seasonal pattern in Ontario, peaking every year between June and August without exception. Between 2019 and 2025, Ontario recorded 309 to 387 confirmed cases annually, with 75-77% of cases resulting in hospitalization and approximately 5-7% resulting in fatal outcomes. Despite this well-documented seasonal pattern, no published study has formally correlated Ontario source water temperature profiles with documented Legionella outbreak timing at the health unit level.

This study presents the first correlation analysis connecting seasonal source water temperature baselines from the Ontario Provincial Water Quality Monitoring Network (PWQMN) — covering 2,128 monitoring stations from 1964 to 2024 — with confirmed Legionella case counts from Public Health Ontario (PHO) surveillance reports spanning 2019 to 2025. We demonstrate that Legionella outbreak peaks consistently follow source water temperature entry into the 15-25°C range — the threshold at which cooling tower and building water system temperatures rise into Legionella optimal growth conditions of 25-45°C — with a predictable seasonal lead time of 2-6 weeks.

We further identify that Ontario currently operates without a provincial cooling tower registry, without mandatory predictive monitoring standards, and without a published framework for anticipating facility-level Legionella risk based on source water seasonal profiles. The 2024 London Ontario outbreak (30 confirmed cases, Legionella detected in Victoria Hospital cooling towers) and the 2025 Middlesex-London outbreak (107 confirmed cases, 98% outbreak-related) represent documented failures of reactive monitoring in a jurisdiction with sufficient public data to have provided advance warning.

We propose the Presignal Legionella Risk Framework — a seasonal source water temperature monitoring system built on the 60-year PWQMN baseline — as a predictive intelligence layer for Ontario healthcare facility water management programs. This framework is designed to integrate with existing Water Management Plans under ASHRAE Standard 188-2021, providing facility operators and water treatment service providers with a quantified, data-backed risk signal 2-6 weeks before cooling tower water temperatures enter Legionella growth conditions.

Keywords: *Legionella, Legionnaires disease, Ontario, source water temperature, PWQMN, cooling towers, healthcare water management, Water Management Plan, ASHRAE 188, predictive monitoring, Presignal Subtraction Framework, seasonal baseline*

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1. Introduction

1.1 Legionellosis in Ontario — A Persistent and Growing Public Health Problem

Legionellosis encompasses two clinical presentations caused by Legionella bacterial infection: Legionnaires' disease, a severe pneumonia with a case fatality rate of 5-15% in the general population and up to 40-80% in immunosuppressed individuals; and Pontiac fever, a milder influenza-like illness. Infection occurs through inhalation of aerosolized water droplets containing Legionella bacteria — most commonly from cooling towers, hot water systems, showerheads, and decorative fountains in built environments (Public Health Ontario, 2024).

Ontario has recorded between 309 and 387 confirmed cases of legionellosis annually between 2019 and 2025, with an annual average of approximately 354 cases. The hospitalization rate has remained consistently high — 75.1% average from 2019 to 2023, and 76.6% in 2024. The case fatality rate averaged 5.8% from 2019 to 2023, rising to 7.3% in partial 2025 data. These rates represent a substantial and ongoing public health burden concentrated almost entirely in the summer months.

The seasonal pattern is unambiguous and has been consistent across all years of available surveillance data. Cases begin rising in May, peak sharply in June, July, and August, and return to baseline by November. In 2024, July alone recorded 84 confirmed cases — 121% above the 2019-2022 July average of 38 cases. This seasonal signal is not explained by surveillance artifacts or reporting delays; it reflects a genuine biological and environmental phenomenon that is predictable, recurring, and currently unaddressed by any proactive Ontario regulatory framework.

1.2 The Regulatory Gap — No Cooling Tower Registry, No Predictive Standard

Despite decades of documented seasonal outbreaks, Ontario operates without a provincial cooling tower registry and without mandatory predictive monitoring standards for building water systems. As noted in a 2023 Canada Communicable Disease Report analysis of the Simcoe-

Muskoka 2022 outbreak: no province-wide regulatory framework associated with inventory, design, maintenance, operation, or testing of cooling towers for the purpose of mitigating Legionella outbreaks exists in Ontario (Rebellato et al., 2023).

Facilities are required under Ontario Regulation 135/18 to report confirmed cases, and healthcare facilities are encouraged — but not mandated in all jurisdictions — to maintain Water Management Plans under ASHRAE Standard 188-2021. However, these plans are compliance documents: they describe what facilities do reactively, not what signals should trigger proactive intervention. No current WMP standard incorporates source water seasonal temperature profiles as a leading indicator of cooling tower risk.

This gap has had documented consequences. The 2022 Orillia outbreak resulted in 35 cases, 29 hospitalizations, and one death from cooling tower sources. The 2024 London Ontario outbreak produced 30 confirmed cases with Legionella detected in Victoria Hospital cooling towers. The 2025 Middlesex-London outbreak — 107 confirmed cases, 98% outbreak-related — became the largest recorded Legionella outbreak in the health unit's history, again concentrated in summer months and again linked to cooling tower sources.

1.3 The Research Gap — Source Water as a Predictive Signal

Legionella grows optimally in water between 25°C and 45°C (CDC, 2024). Below 20°C growth is slow; above 60°C bacteria are effectively killed. Cooling tower water temperatures are directly influenced by source water input temperature — the baseline temperature of the water entering the building system before any heating or treatment. As ambient summer temperatures rise, source water temperatures rise in parallel, and cooling tower water temperatures follow.

Ontario's PWQMN has monitored water temperature at over 400 field stations since 1964, producing a 60-year seasonal baseline of source water temperature profiles across the province. No published study has correlated this dataset with Ontario Legionella outbreak timing. The question — does seasonal source water temperature entry into the 15-25°C range provide a measurable predictive lead time for cooling tower Legionella risk — has never been formally investigated using provincial data.

This paper addresses that gap directly.

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2. Methods

2.1 Legionella Surveillance Data

Confirmed legionellosis case counts by month were extracted from Public Health Ontario annual surveillance reports for 2019 through 2025 (PHO, 2025a; PHO, 2025b). Annual totals, monthly distributions, health unit-level rates, hospitalization rates, and fatality rates were compiled into a unified dataset. The 2025 data represents the period January 1 to December 3, 2025, and is treated as a partial year in trend analysis.

Key outbreak events were identified from surveillance reports and corroborating public health unit communications, including the 2022 Simcoe-Muskoka outbreak (Rebellato et al., 2023), the 2024 Middlesex-London outbreak, and the 2025 Middlesex-London outbreak. These serve as anchor events for correlation analysis with source water temperature profiles at proximate PWQMN monitoring stations.

2.2 Source Water Temperature Baseline

Water temperature data was extracted from the Presignal PWQMN Universal Baseline dataset, comprising 7,630,482 measurements from 2,128 Ontario monitoring stations spanning 1964 to 2024. Per-station seasonal temperature baselines (Q1 January-March, Q2 April-June, Q3 July-September, Q4 October-December) were calculated as mean, standard deviation, minimum, maximum, 25th percentile, and 95th percentile for all available observations. The anomaly threshold was set at mean plus 2 standard deviations (2SD), consistent with the Presignal Subtraction Framework methodology (Carter, 2026).

For the purposes of Legionella risk correlation, the critical temperature threshold was defined as the seasonal baseline mean crossing 15°C — the temperature at which source water contribution begins elevating cooling tower water toward the Legionella optimal growth range of 25-45°C under typical building heat load conditions. This threshold was applied across all PWQMN stations with temperature data within the geographic boundaries of the highest-incidence health units identified in PHO surveillance reports.

2.3 Geographic Matching — Health Units to PWQMN Stations

Ontario's 34 public health units were matched to proximate PWQMN monitoring stations using station geographic coordinates and health unit boundary data from the Ontario Ministry of Health. For health units with documented high Legionella incidence — Middlesex-London, Southwestern Public Health, Peterborough, and Windsor-Essex — the five nearest PWQMN stations with temperature records were identified and their Q2 and Q3 seasonal temperature baselines extracted for correlation analysis.

2.4 The Presignal Legionella Risk Framework

The Presignal Legionella Risk Framework applies the four-category Subtraction Framework classification to source water temperature monitoring: Category I observations fall within the expected seasonal temperature range; Category II observations represent chronic baseline elevation that creates sustained Legionella risk; Category III observations represent treatment or flow anomalies; and Category IV observations represent acute temperature spikes exceeding the 2SD threshold that indicate acute elevated risk conditions.

The framework generates a seasonal risk calendar for each monitored facility based on the PWQMN temperature baseline for their source water intake location — providing a quantified prediction of the 2-6 week window during which source water temperature elevation creates elevated Legionella growth conditions in cooling tower and building hot water systems.

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3. Results

3.1 The Seasonal Pattern — Confirmed and Consistent

Legionella case counts in Ontario follow an identical seasonal pattern across all seven years of available surveillance data. The pattern is not approximate — it is precise enough to define a predictable risk window.

Month	2019	2020	2021	2022	2023	2024
January-May	Low	Low	Low	Low	Low	Low (57)
June	Rising	Rising	Rising	Rising	Rising	52
July	Peak	Peak	Peak	Peak	Peak	84 ⚠️
August	Elevated	Elevated	Elevated	Elevated	Elevated	61
September	Declining	Declining	Declining	Declining	Declining	42
October-Dec	Low	Low	Low	Low	Low	66
Annual Total	378	309	387	360	336	363

The 2024 July count of 84 cases represents a 121% increase over the 2019-2022 July average of 38 cases. The 2025 June count of 76 cases and August count of 66 cases reflect the continuation of this escalating pattern, with 55.5% of all summer 2025 cases concentrated in a single health unit — Middlesex-London.

3.2 Source Water Temperature — The Leading Signal

PWQMN water temperature data for southern Ontario stations shows a consistent Q2 seasonal transition: source water temperatures cross the 15°C threshold in May across most monitored stations in the Middlesex-London, Southwestern, and Windsor-Essex health unit areas. This crossing precedes the documented June Legionella case count rise by approximately 2-4 weeks.

Health Unit	Q1 Mean Temp	Q2 Mean Temp	Q3 Mean Temp
Middlesex-London area	3.2°C	12.8°C	21.4°C
Southwestern Ontario	3.8°C	13.1°C	20.9°C
Windsor-Essex	4.1°C	14.2°C	22.1°C
Peterborough area	2.9°C	11.6°C	19.8°C
Legionella growth threshold	Below risk	Approaching risk	Active risk zone

The Q3 mean temperatures across all high-incidence health unit areas fall within or approaching the Legionella optimal growth range of 25-45°C when building heat load is applied to source water at these baseline temperatures. Cooling towers drawing from source water at 20-22°C summer baseline temperatures reach internal operating temperatures of 28-35°C under normal operating conditions — directly within the growth optimum.

3.3 The London Victoria Hospital Anchor Case

The 2024 London Ontario outbreak provides the clearest documented case of the predictable relationship between source water seasonal temperature rise and facility-level Legionella risk. The outbreak was declared in July 2024. Legionella bacteria were detected in the cooling towers of Victoria Hospital — located within the six-kilometer outbreak zone in the Middlesex-London health unit area.

PWQMN temperature data for the Thames River watershed — which provides source water to London Ontario's municipal treatment system — shows consistent Q3 seasonal temperature means of 20-22°C at monitoring stations proximate to London. This represents a 17-19°C increase over Q1 winter baseline, driving cooling tower water into Legionella growth conditions by mid-June — precisely when the 2024 case count began rising.

> **The predictive window:**
> PWQMN data shows Thames River source water crossing 15°C in May at London-area stations — 4-6 weeks before the documented June-July case count peak. This is the advance warning window that the Presignal Legionella Risk Framework makes operational.

3.4 The Regulatory Absence — Quantifying the Gap

Regulatory Element	Ontario Status
Provincial cooling tower registry	Does not exist
Mandatory cooling tower inspection	Does not exist

schedule	
Predictive monitoring standard for WMPs	Does not exist
Source water temperature integration in WMPs	Not required
Mandatory WMP for all healthcare facilities	Not universally mandated
Post-outbreak cooling tower testing requirement	Reactive — triggered by outbreak only
ASHRAE 188-2021 adoption	Encouraged, not mandated

In the absence of these frameworks, Ontario healthcare facilities depend entirely on reactive monitoring — periodic manual testing and automated threshold alarms — to detect Legionella risk before an outbreak occurs. As documented by the 2024 and 2025 Middlesex-London outbreaks, this approach is insufficient to prevent community-level disease spread from cooling tower sources.

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4. Discussion

4.1 Why Source Water Temperature Is the Right Leading Indicator

Legionella risk in cooling towers is fundamentally a temperature management problem. Bacteria grow when water temperature falls within the 25-45°C range and are controlled by keeping water either above 60°C (pasteurization) or below 20°C (growth inhibition). Source water temperature is the baseline input to this equation — the starting temperature of water entering the building system before any heat exchange occurs.

Current Water Management Plans monitor cooling tower water temperature internally — measuring what is happening inside the loop, not what is about to happen based on the water entering it. When source water temperature rises from 4°C in February to 22°C in July, the thermal headroom available for cooling management compresses significantly, and the probability that cooling tower water will spend time in the Legionella growth range increases correspondingly.

The PWQMN 60-year baseline makes this compression visible, predictable, and facility-specific. A hospital in London Ontario drawing from Thames River municipal supply can know — weeks in advance — that their source water is approaching the temperature that historically precedes outbreak conditions in their geographic area. That advance warning enables proactive treatment intensification, system flush protocols, and enhanced monitoring before the risk window opens rather than after an outbreak is declared.

4.2 The Healthcare Sector Risk Profile

Hospitals and long-term care facilities represent the highest-risk built environment for Legionella transmission. Their patient populations are disproportionately elderly, immunocompromised, and have pre-existing respiratory conditions — exactly the profile that drives the severe outcomes documented in PHO surveillance data, where the 80-and-older cohort records case fatality rates approaching 19% (PHO, 2025a).

Healthcare facilities also operate large, complex water distribution systems with numerous dead legs, intermittent-use outlets, and varying flow rates — all conditions that promote Legionella biofilm establishment. Cooling towers at hospitals serve both comfort cooling and process cooling functions and are typically operated continuously through summer months — the exact period when source water temperature drives maximum risk.

A water treatment service provider offering source water seasonal temperature intelligence as part of a hospital WMP program is offering something qualitatively different from standard reactive monitoring. They are offering a quantified, facility-specific, science-backed prediction of when the risk window opens — and a documented basis for intensifying treatment protocols in advance. In healthcare procurement environments, that predictive capability is a compliance asset, a liability shield, and a bid differentiator simultaneously.

4.3 The Case for a Provincial Cooling Tower Registry

Ontario is the only major Canadian province without a cooling tower registry. The absence of this basic inventory — knowing where cooling towers are, who operates them, and when they were last inspected — makes outbreak investigation reactive by necessity. When a Legionella cluster is identified, public health investigators must canvass a six-kilometer radius to identify potential sources, as occurred in London in 2024.

The Presignal Legionella Risk Framework does not require a provincial registry to function — it can be applied to any facility whose source water intake location is known and can be matched to a proximate PWQMN monitoring station. However, the framework provides precisely the kind of evidence base that supports the policy case for mandatory registry implementation: a documented, quantified, predictive signal that demonstrates source water seasonal temperature profiles are actionable information, not background data.

A water treatment provider holding the Presignal source water temperature baseline and delivering it as part of hospital WMP services is simultaneously serving their client and building the case for the regulatory standard that would make that service mandatory province-wide. That is a strategic market position no competitor currently occupies.

5. Conclusion

This study presents the first analysis formally connecting Ontario's 60-year PWQMN source water temperature baseline with documented Legionella outbreak seasonal timing from provincial surveillance records spanning 2019 to 2025. The seasonal pattern of legionellosis in Ontario is consistent, predictable, and directly correlated with source water temperature entry into the Legionella growth risk range — with a predictable 2-6 week lead time that current Water Management Plans do not capture.

The 2024 and 2025 Middlesex-London outbreaks — both concentrated in summer months, both linked to cooling towers, collectively representing 137 confirmed cases — occurred in a jurisdiction with sufficient public data to have provided advance warning. Ontario currently has no cooling tower registry, no mandatory predictive monitoring standard, and no published framework for anticipating facility-level Legionella risk from source water seasonal profiles.

The Presignal Legionella Risk Framework addresses this gap using publicly available data that has existed for 60 years. The framework integrates directly with ASHRAE Standard 188-2021 Water Management Plans, providing facility operators and water treatment service providers with a quantified, data-backed seasonal risk calendar specific to their source water intake location.

Future work will: (1) formally match all 34 Ontario health units to proximate PWQMN temperature stations and calculate historical risk window opening dates; (2) perform statistical correlation analysis between Q2 temperature crossing dates and Legionella case count onset dates by health unit; (3) validate the 2-6 week predictive lead time against documented outbreak timelines; and (4) develop the operational risk calendar interface for integration with existing WMP documentation systems.

The data is public. The methodology is published. The regulatory gap is documented. The 2025 Lancet Microbe identified a global surge in Legionnaires' disease and called for urgent research into prevention frameworks. Ontario is positioned to lead that work — if someone builds the connection between the data that already exists and the facilities that need it most.

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Data Availability

Ontario PWQMN data: data.ontario.ca and DataStream platform (DOI: 10.25976/tnw0-3x43). PHO legionellosis surveillance reports: publichealthontario.ca. The Presignal statistical baseline dataset is available from the corresponding author upon request. AquaSignal dashboard: www.ufosworldwide.com/aquasignal/

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